

Underfitting or Overfitting ?

Analyse the curves

Analyse the curves



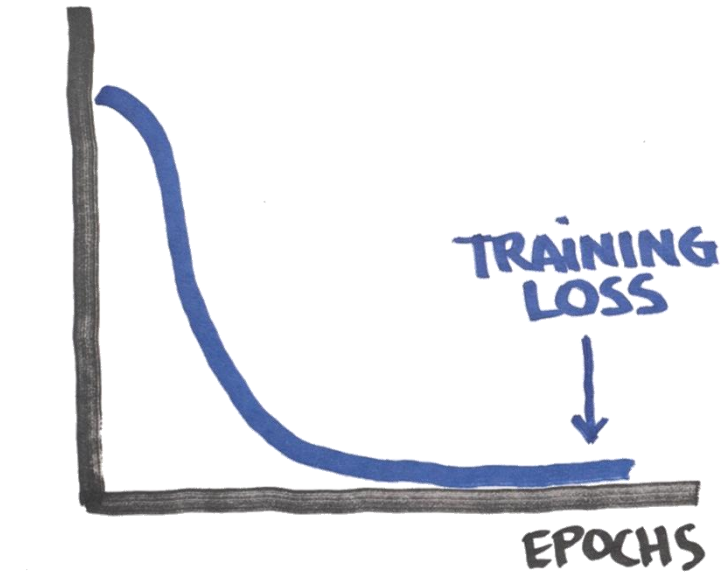
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Analyse the curves



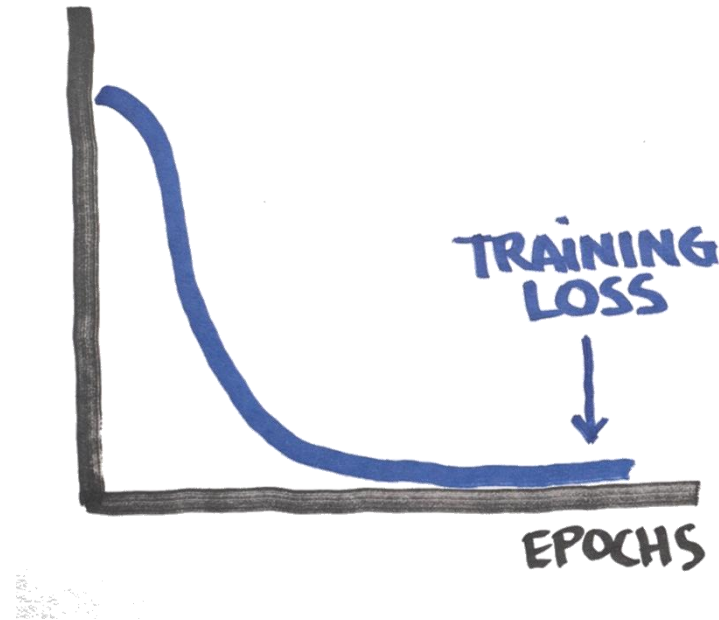
- Model is maybe underfitting, but we cannot say

Analyse the curves



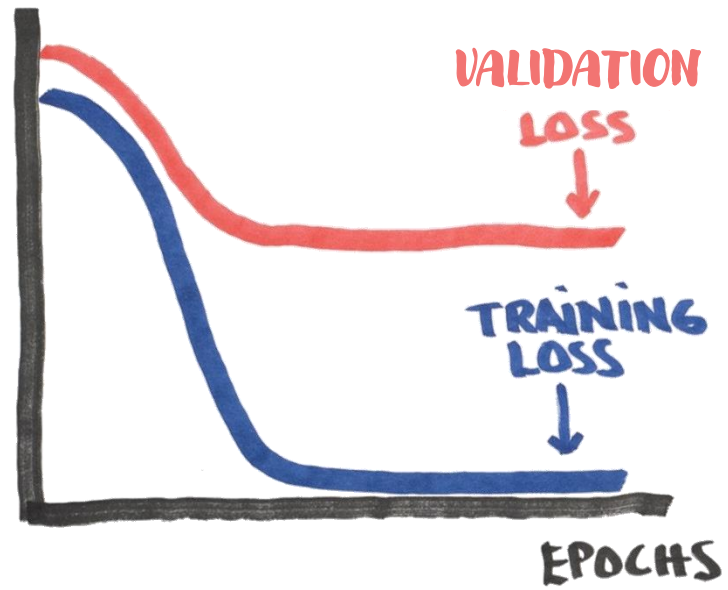
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Analyse the curves

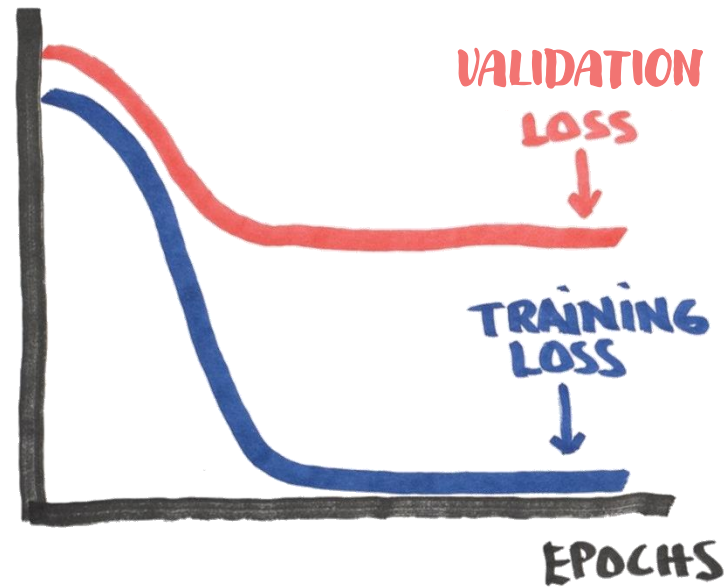


- Model is maybe overfitting, but we cannot say
- We need a validation curve

Analyse the curves

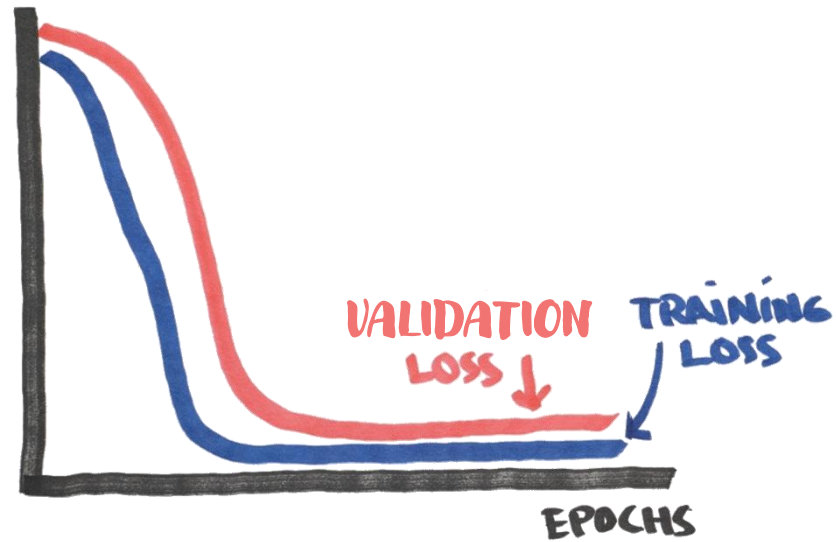


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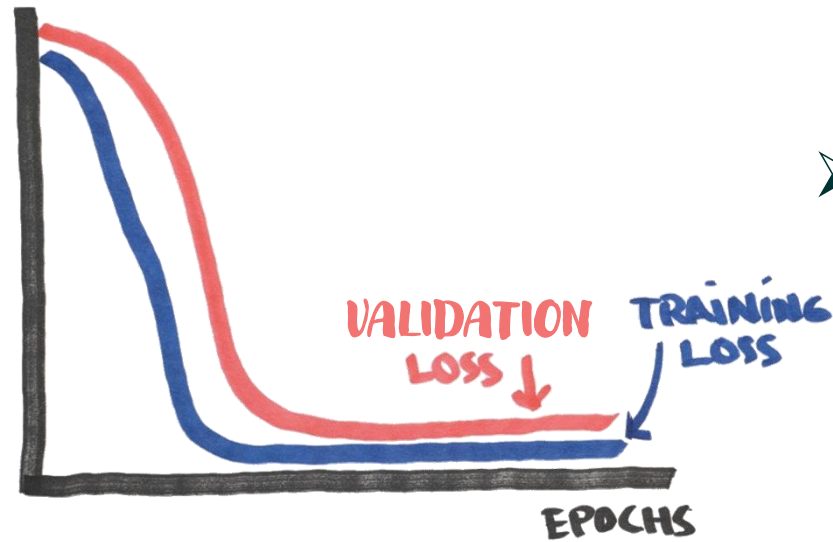


- Model is overfitting
- See actions versus overfitting

Analyse the curves

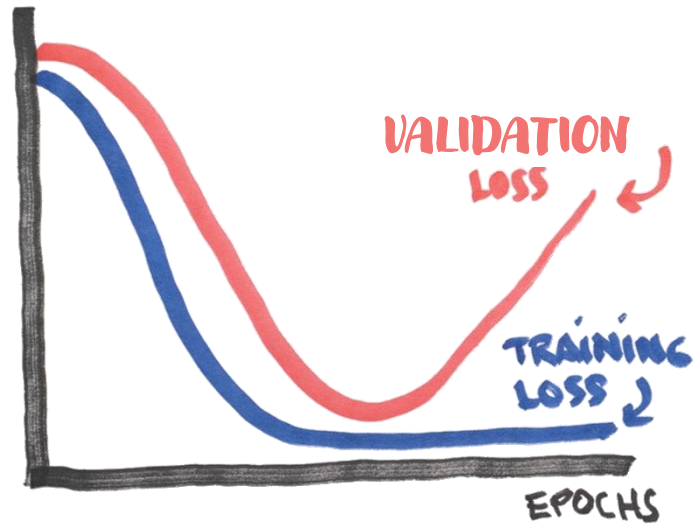


Analyse the curves

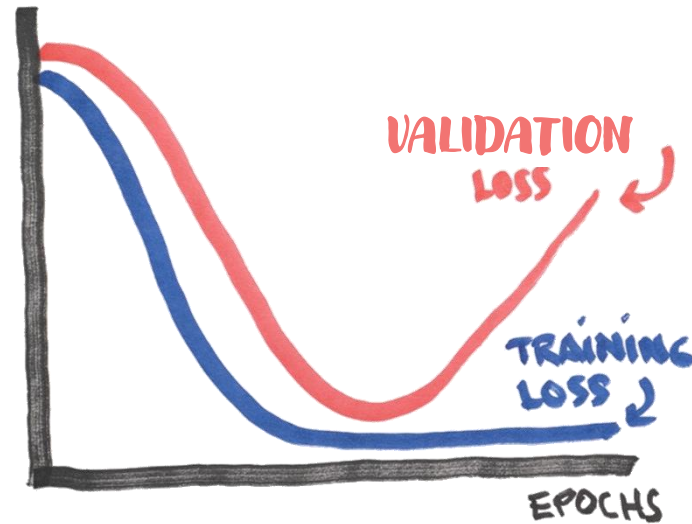


➤ Model is performing well!

Analyse the curves

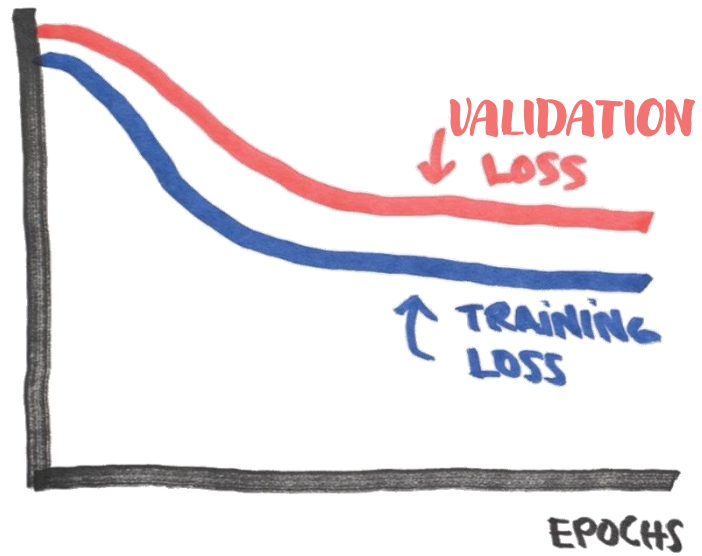


Analyse the curves

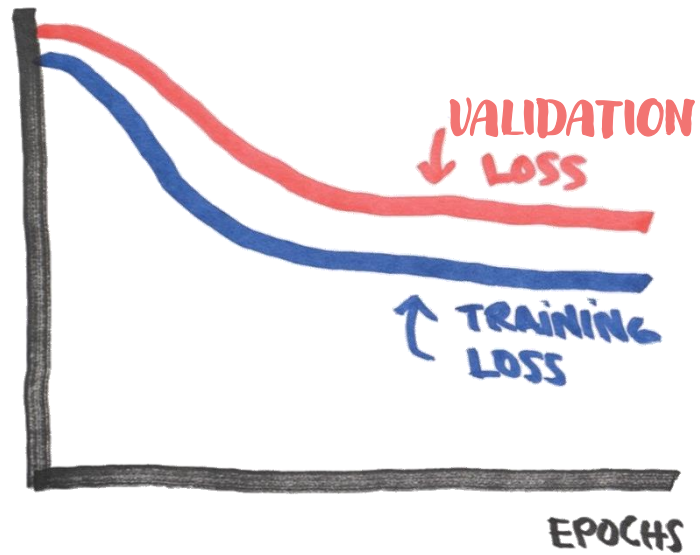


- Model is training well, then it overfits
- Use early stopping

Analyse the curves



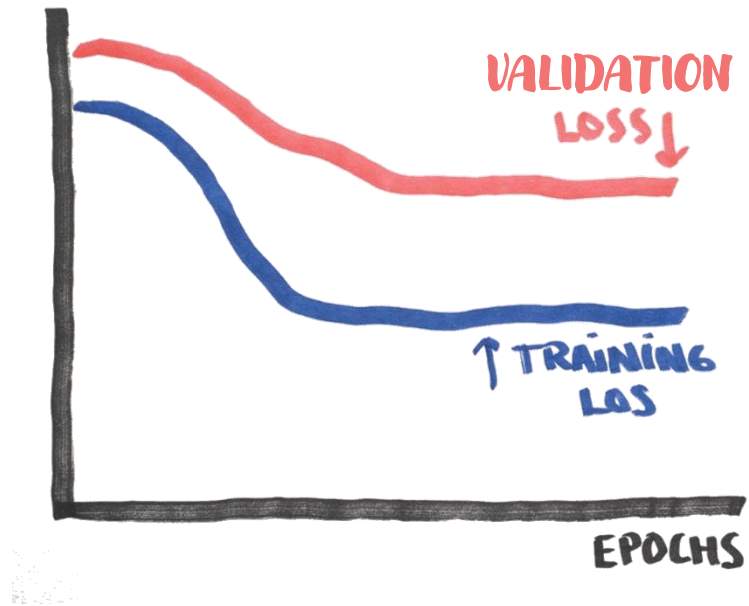
Analyse the curves



➤ Model is underfitting

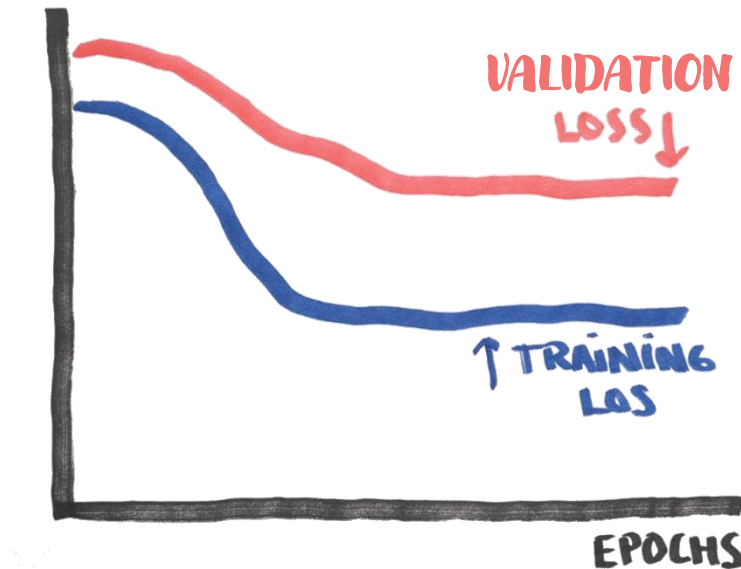
➤ See actions versus underfitting

Analyse the curves



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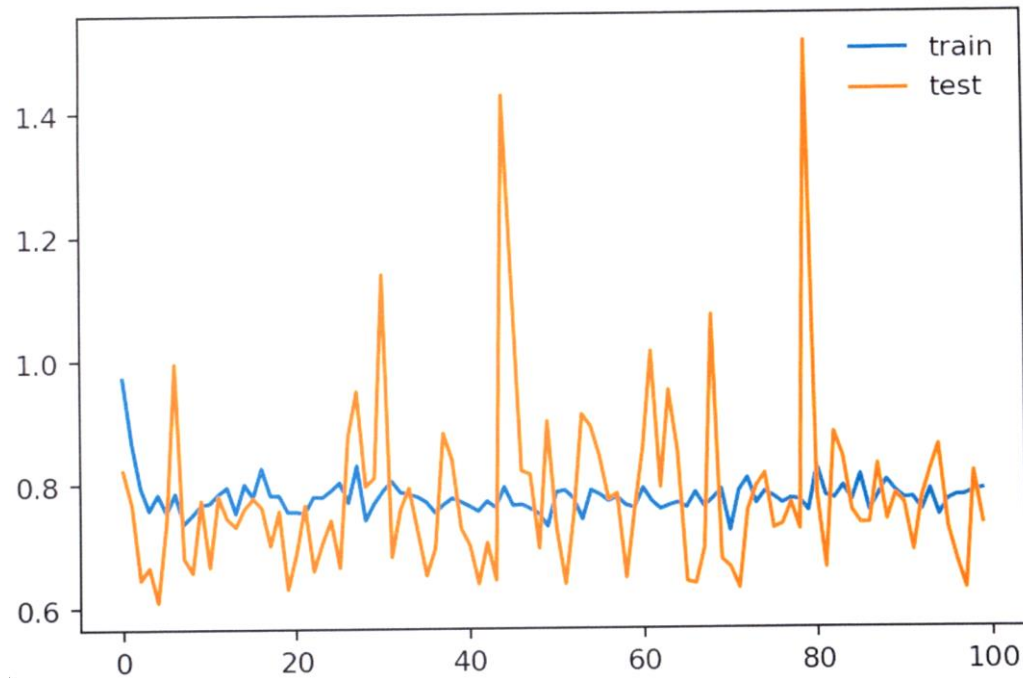
Analyse the curves



- Hard to say!
- Model may be slightly underfitting
- Train dataset may not cover the same distribution that test dataset
 - Improve / add training data

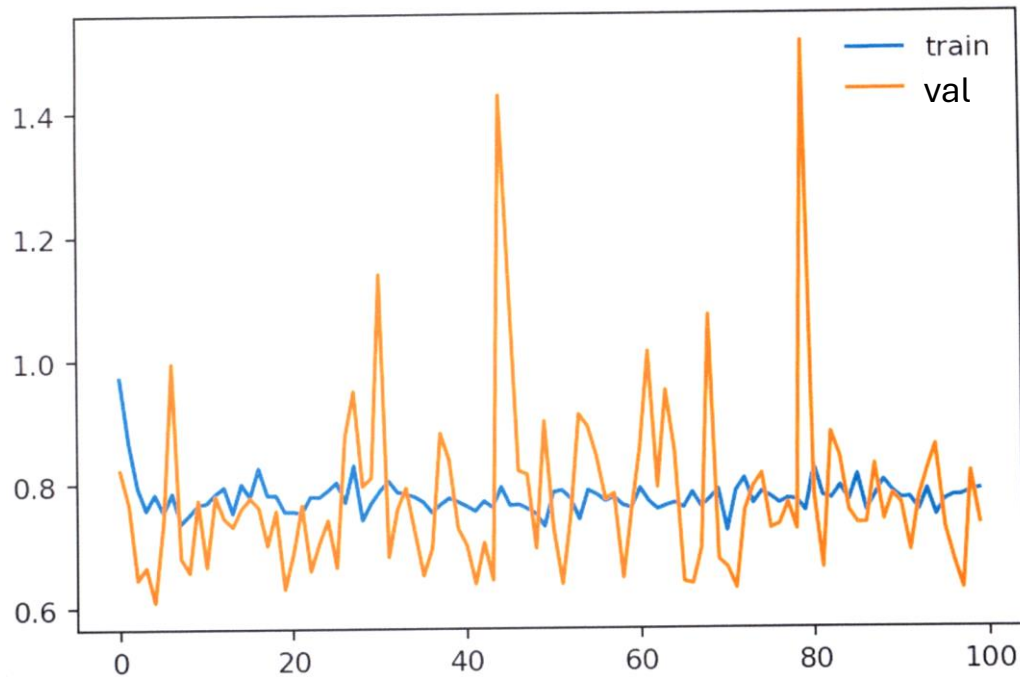
Analyse the curves: What is the problem, how to fix it?

using mini-batch gradient descent to train the network:



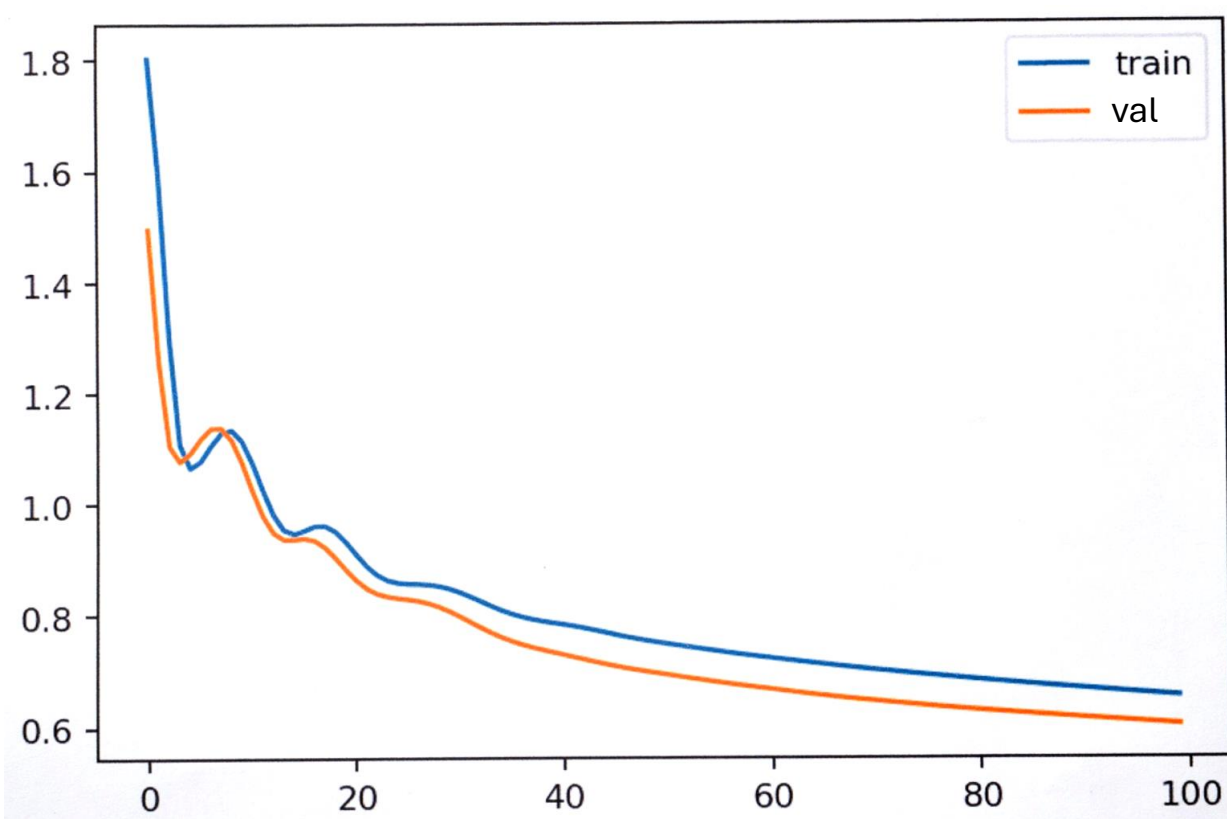
Analyse the curves

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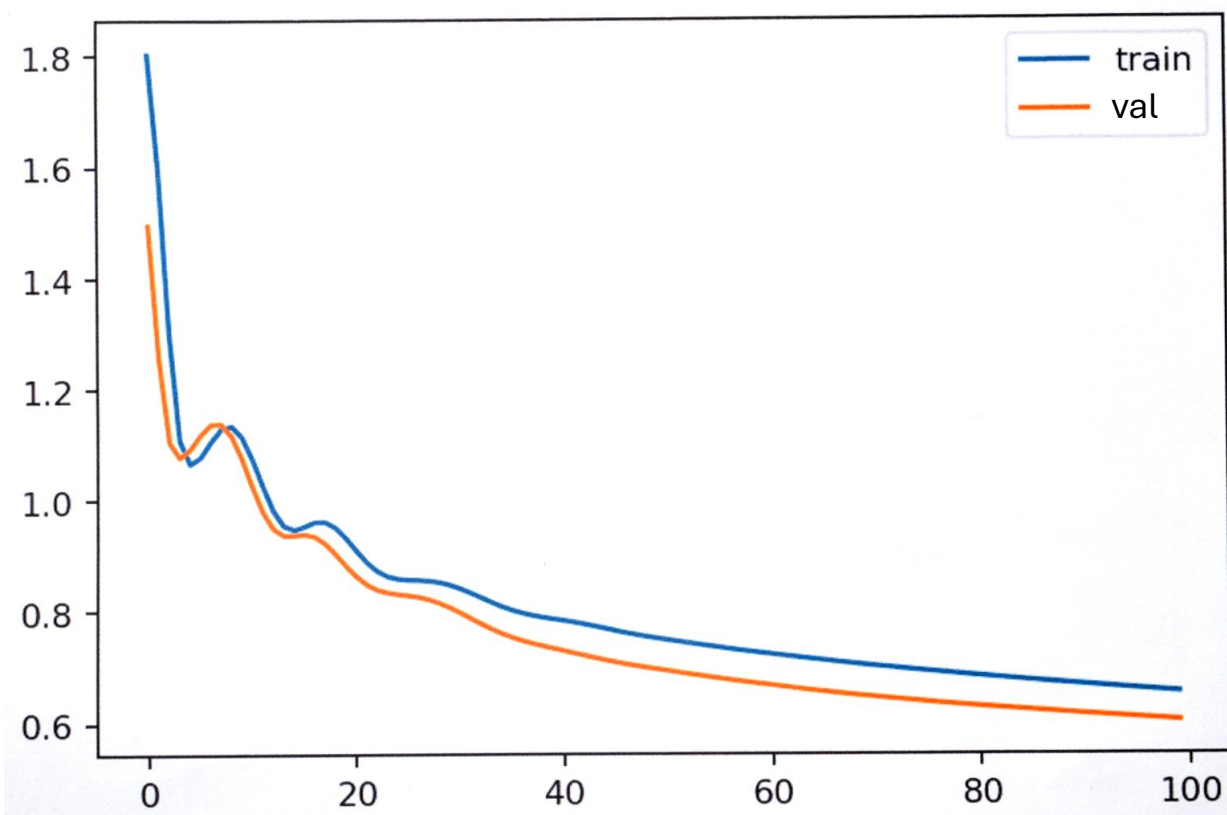
- A learning rate that's too high could be causing the oscillation
- In a small batch, one instance could swing the loss significantly
- Use a lower learning rate
- Increase batch size to smooth the loss

Analyse the curves



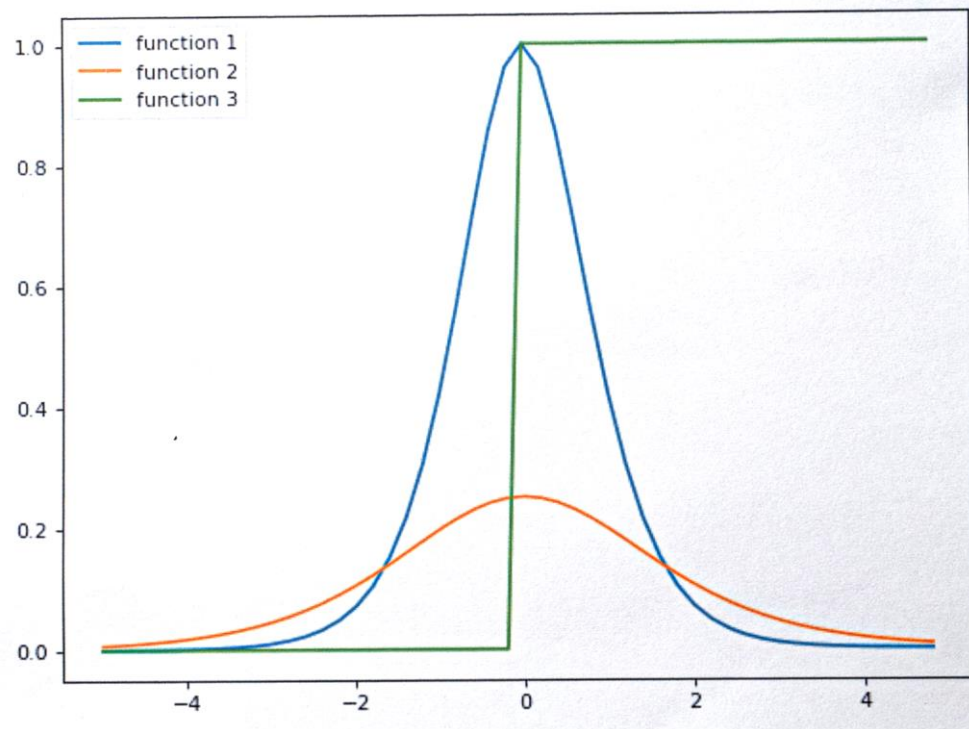
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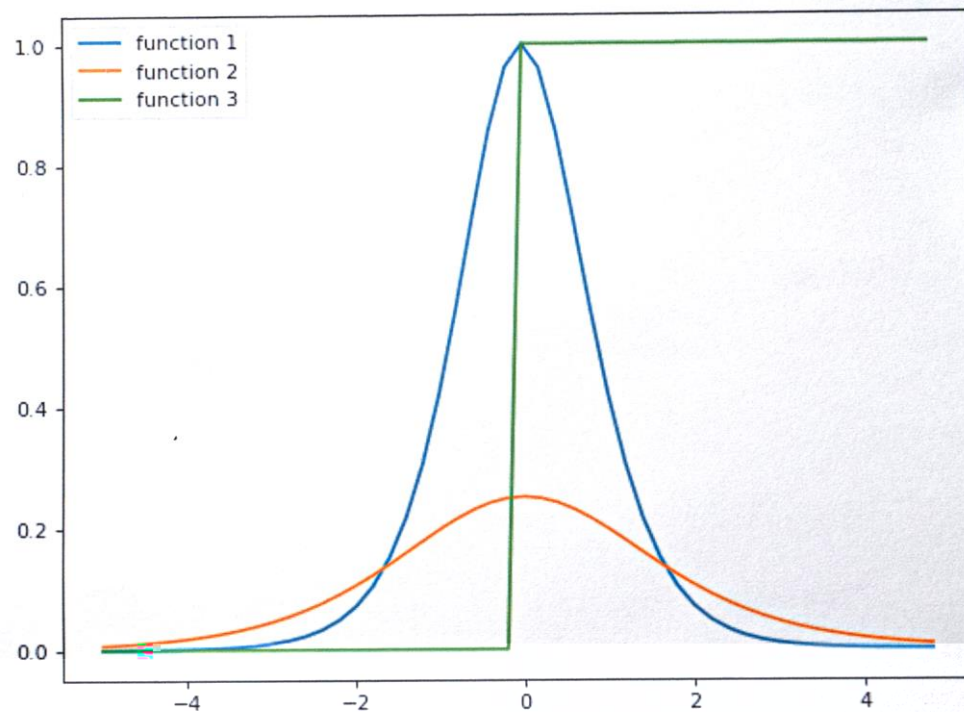
➤ The network has no problem!

From which functions are these gradients?



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From which functions are these gradients?



1. TanH
2. Sigmoid
3. ReLu